MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2023/2024

SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE IN AGRICULTURAL ECONOMICS, BACHELOR OF SCIENCE IN AGRIBUSINESS MANAGEMENT, BACHELOR OF SCIENCE IN ANIMAL SCIENCE AND BACHELOR OF SCIENCE IN FISHERIES AND AQUACULTURE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

AEG 204: PRODUCTION ECONOMICS

Date: 25th April, 2024 Time: 8.30 - 11.30 am

INSTRUCTIONS:

- DO NOT write anywhere on this question paper
- Answer ALL questions in SECTION A and any other TWO from SECTION B

10/6 3:30 (120) 25/4 8:30

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SECOND YEAR SECOND SEMESTER FOR THE DEGREE OF BACHELOR OF SCIENCE IN AGRICULTURAL ECONOMICS/ AGRIBUSINESS MANAGEMENT/ ANIMAL SCIENCES/ FISHERIES WITH IT

AEG 204: PRODUCTION ECONOMICS

DATE: **TIME:** 3 HOURS

Instructions:

1. Answer **ALL** the questions in Section **A** and Any **TWO** questions in Section B.
2. Marks for each question are indicated in brackets against each question

SECTION A (COMPULSORY) [40 Marks]

Question 1

a) From the table given below, find:

Output units	0	1	2	3	4	5	6
Total Cost	600	900	1000	1050	1150	1350	1500
Total Fixed cost	600	600	600	600	600	600	600

(i) Total Variable Cost. (2 marks)

(ii) Average Fixed Cost. (2 marks)

(iii) Average Variable Cost. (2 marks)

(iv) Average Total Cost. (2 marks)

(v) Marginal Cost. (2 marks)

b) Draw a graph of MC, AVC, AFC, and ATC against Y. (2 marks)

Question 2

Given the following production function

$$Y = 4X + 0.06X^2 - 0.001X^3$$

a) Complete the table below. (8 marks)

Input (X)	Output (Y)	Average Physical Product APP	Marginal Physical Product (MPP)	Elasticity of Production (E_P)
0				
5				
10				
15				
20				
25				
30				
35				
40				
45				
50				
55				
60				
65				
70				
75				
80				

b) Plot the TPP, APP, and MPP curves. Show the 3 stages of production. (5 marks)

c) Briefly discuss each region in (b) above and explain what a rational producer should do in each region. (5 marks)

Question 3

a) The following are input combinations of X_1 and X_2 that can produce 20 units of output. The price of X_1 is Kshs 200 and that of X_2 is Kshs.300.

Output (Y)	X_1	X_2
20	0	20
20	8	16
20	16	12

20	24	8
20	32	4
20	40	0

i) What factor-factor relationship do the two inputs exhibit? (2 marks)

ii) Determine the optimal input combination. (4 marks)

b) Suppose the price of X_1 is Kshs 250 and the price of X_2 is Kshs 220, determine the new optimal input combination. (4 marks)

SECTION B [30 MARKS]

Question 4

a) Agricultural production is faced with lots of risks and uncertainties, discuss the main sources of risks and uncertainties in agricultural production. (7 marks)

b) Discuss ways of mitigating the above risks and uncertainties in agricultural production. (8 marks)

Question 5

a) With the help of the production function

$$Y = 100 - 3X_1^2 + 4X_1 + 2X_1X_2 - 5X_2^2 + 48X_2$$

Find the value of X_1 and X_2 at which Total Physical Product is at maximum. (6 marks)

b) Discuss land as a production resource managers have to deal with wisely under the following subheading

i) Quality. (3 marks)

ii) Availability. (3 marks)

iii) Flexibility. (3 marks)

Question 6

a) Consider the production function $Y = 70 + 2X - 0.02X^2$

i) Find the level of X at which Y is at maximum. (4 marks)

ii) Find levels of X maximizing net returns when $P_X = 1$ and $P_Y = 10$. (3 marks)

b) Using a suitable diagram, discuss the four types of product-product relationships commonly found in Agriculture. (8 marks)